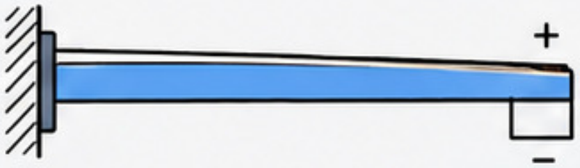
A Why Endpoint-Only Evaluation Fails

Example: Piezoelectric Cantilever Design



Design variables:
L, w, t, material,
layers, V_{drive}, ...

1 Triage
Is the request actionable?
Clarify intent, constraints, outputs.

2 Repair/Search
Use verifier feedback.
Iterate designs, fix issues, close gaps.

3 Recover
Bounce back after failure.
Detect failure, re-anchor, resume search.

4 Rank
Choose among feasible options.
Compare feasible candidates and select best.

One-Score Benchmark (Endpoint-Only)
Aggregates only the final answer.
Masking failures in process and control.

A model can rank feasible designs well yet still fail to find or recover them.

Workflow Risks

- wrong go/no-go
- poor closure
- state contamination
- policy mismatch

Why it matters

- hard physical constraints
- verifier grounding
- deployment realism
- role-specific failure modes

B VEH-Bench Construction

1 Source & Anchors

209-paper audit

clean design anchors

2 Oracle-Verified Task Builder

$$y = \frac{PL^3}{3EI}$$

shared analytical oracle

feasibility + objective checks

3 Four Diagnostic Probes

- P1 Triage
- P2 Repair
- P3 Recovery
- P4 Ranking

4 Evaluation & Audit

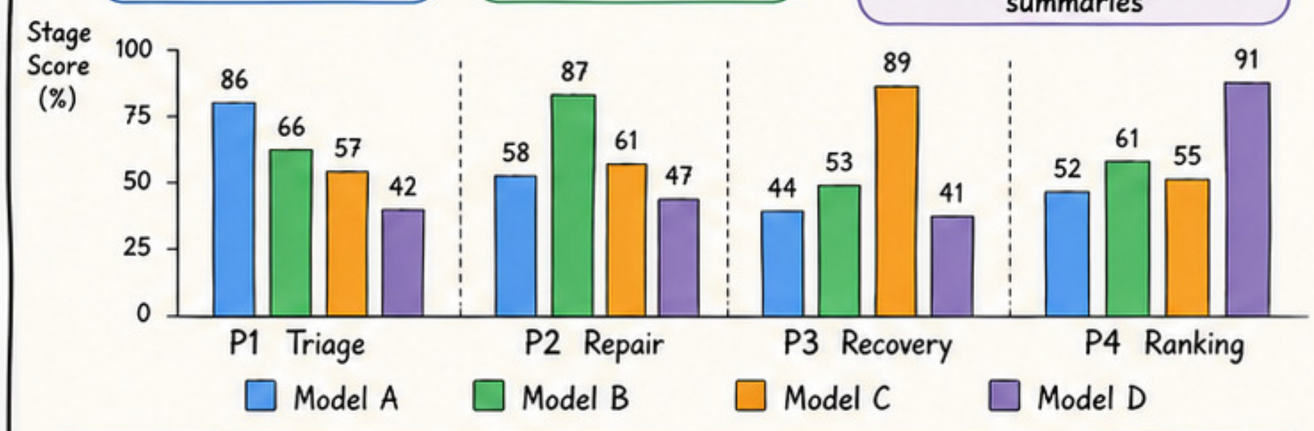
stage metrics • uncertainty reporting • human audit

C What VEH-Bench Reveals

No single model dominates all stages

Best finder ≠ best ranker

Recovery improves when corrupted history is replaced by verifier-authored state summaries



Engineering-agent readiness is a trusted-state control problem, not a single endpoint score.